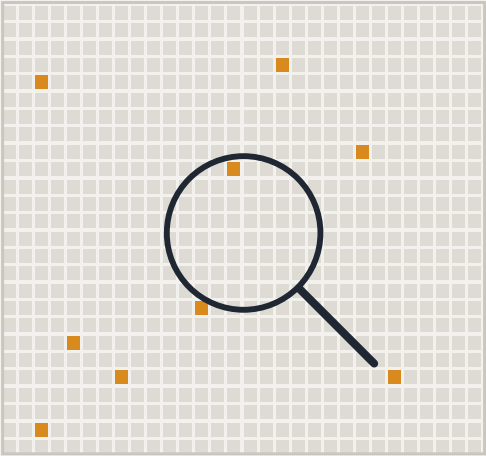


From 6K unlabeled SAE atoms to the five directions a probe really uses — ranked by verified counterfactual flips, then fixed

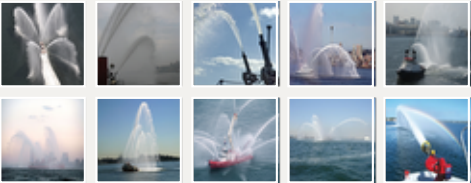
(a) Today: browse an SAE

manual · correlational · not actionable



6,144 atoms — which ones does the fireboat-vs-lifeboat probe use?

per atom: highest-activating images

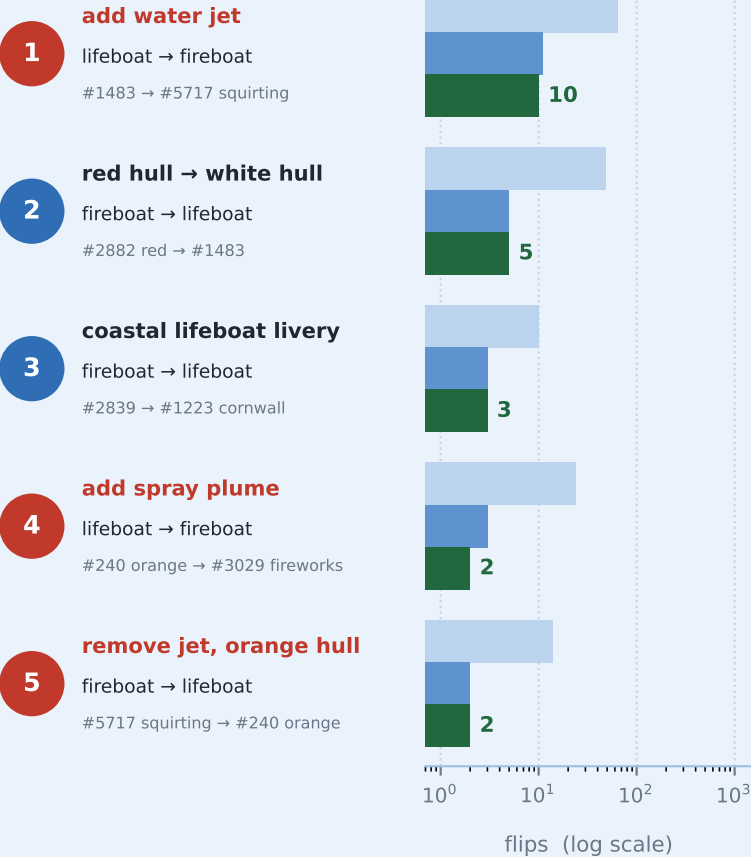


atom #5717 “squirting”: 10 highest-activating training images — all fireboats spraying

✗ jet or boat? unclear · no fix

(b) Rank all directions by verified flips

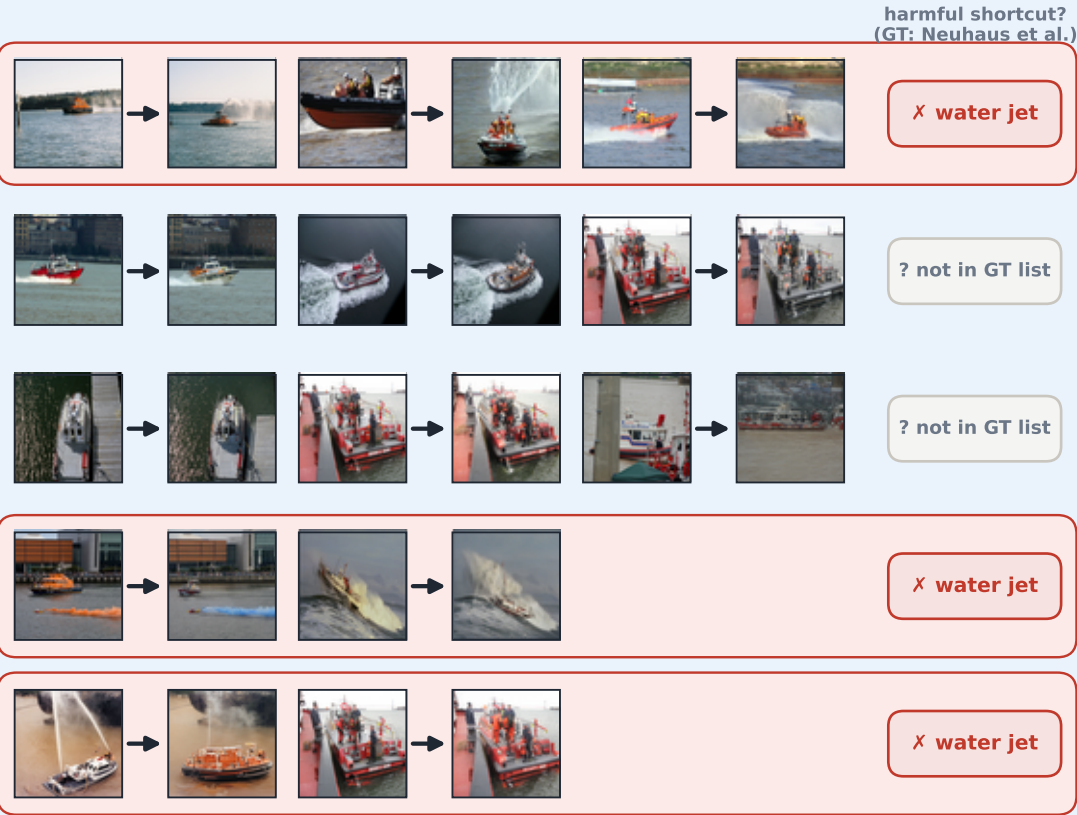
DINOv3 ViT-L/16 linear probe, ImageNet fireboat vs lifeboat (99.2 % test acc.)  
public MSAE on CLIP ViT-L/14 (6,144 atoms), no SAE training  
500 images swept · 313 candidates rendered (DDPM inversion + CFG 2) · 78 directions



- latent flips (distilled probe)
- ambient flips (real probe on decoded image)
- verified flips — rank key

(c) Counterfactuals of the top-5 only (3 verified flips per direction)

original  $\mathbf{x}$  → counterfactual  $\tilde{\mathbf{x}} = \text{decode}(\mathbf{z} + \delta \mathbf{v}_k)$  — the probe's decision flips on every pair

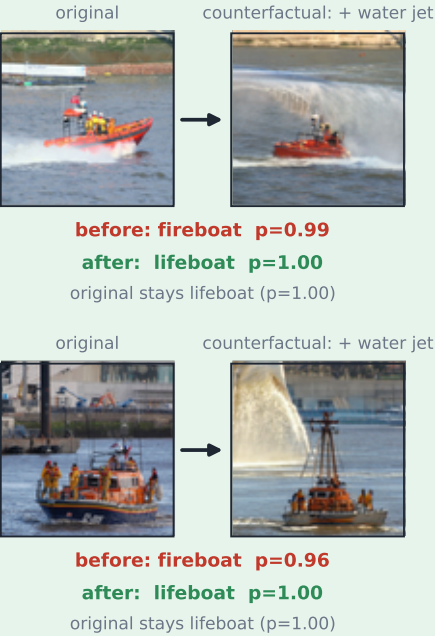


harmful shortcut?  
(GT: Neuhaus et al.)

rank 1, 4 and 5 are the “water jet” shortcut of Neuhaus et al. (2022) for fireboat, rendered as minimal edits of the input: the same lifeboat scene gains a spraying jet and flips to fireboat at  $p = 0.99$ , a fireboat loses its jet and flips to lifeboat — found from the probe alone, with counterfactuals instead of highest-activating samples, and ranked among all directions in a single pass (all shown pairs of a direction when fewer than 3 verified flips exist)

(d) Fix the probe

CFKD on the  $\mathbf{x}$  direction (LLM teacher):  
held-out lifeboat + jet, before → after



52 held-out + train counterfactuals:  
51 fooled the probe before, 1 after  
test acc. 99.2 % → 99.2 %

✓ backbone untouched